CHAPTER 1: CLASSIFICATION OF MATERIALS

A day at the park

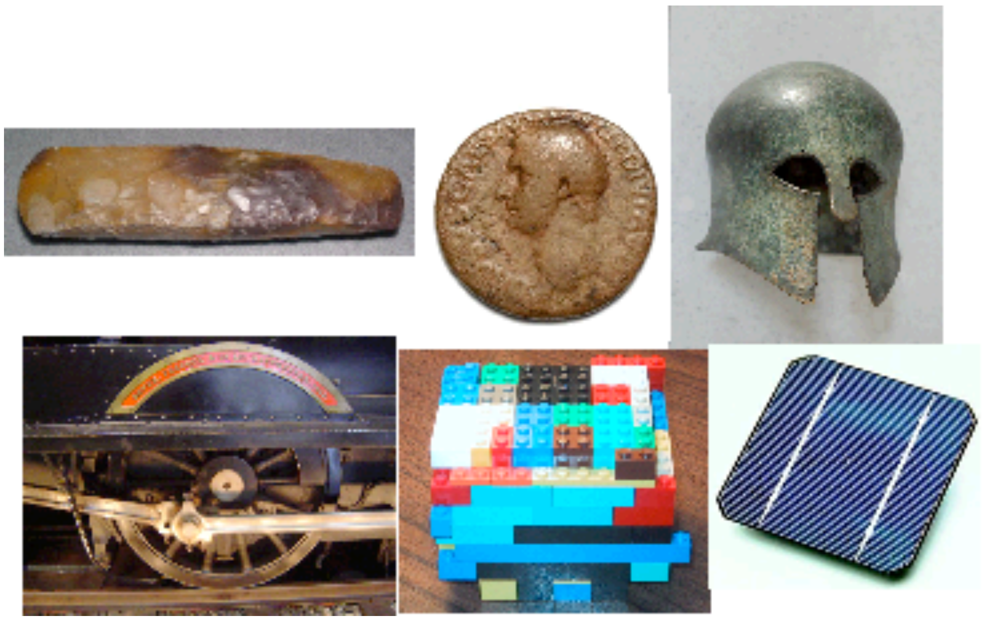
1.1 WHY STUDY MATERIALS?

On a sunny day at the park, where would you prefer to sit and relax - on a wooden bench, stone pedestal, concrete sidewalk, or grassy lawn? It probably depends on if you want to be warm or cool. If you want to read, be thankful that you don't have to carry carved stone tablets or a parchment scroll; you can carry a paperback book in your backpack or download an ebook to a portable computer. Maybe you should take a jacket along. Which would be the best choice - a wool blazer, nylon windbreaker, microfiber trenchcoat, or cotton sweatshirt? People who are fortunate enough to have all of these in their closets can choose the material that best protects them from the predicted weather - temperature, precipitation, and wind.

In modern society we are surrounded by an amazing variety of **materials**. Most of the materials discussed in this book are solids that have been modified from their natural states to make them more suitable for practical applications.

The materials people use have such an impact on their lifestyles that historical eras have been named for them. Ancient artifacts found by archaeologists have been dated and analyzed to reveal the increasing sophistication of their manufacturing methods.

Early humans	**Stone age**
~9000 BCE	**Copper age**
~3000 BCE	**Bronze age**
~1200 BCE	**Iron age**



Artifacts of civilization
top row: flint handaxe, copper coin, bronze helmet
bottom row: locomotive wheel, plastic toy, solar cell

Historians have shown that technological advancements created new tools for agriculture and new weapons for armies. Explorers established trade routes to redistribute raw materials and finished products.

Modern culture is also influenced by the availability of new materials. In the 1960's plastics were used to make colorful toys and housewares at such a low cost that they were frequently disposed of and replaced with the latest style. In the 1980's silicon based electronics started spreading through businesses and homes. Since the 1960s homes (and landfills) in the United States have become bigger and more crowded with objects for applications never dreamed of by stone age humans.

In the 21st century there is much discussion of "globalization." Materials definitely follow a global cycle. Raw materials are collected; processed into useful materials; sold to consumers; and eventually discarded as waste. Each stage may occur on a different continent! Supply and demand of materials can affect international relations.

1.2 WHY STUDY THE CHEMISTRY OF MATERIALS?

A standard place setting includes metal cutlery, a polymer napkin, and a ceramic dish.

Traditionally the three major classes of materials are **metals**, **polymers**, and **ceramics**. Examples of these are steel, cloth, and pottery. These classes usually have quite different sources, characteristics, and applications.

Chemists have learned that the macroscopic (visible) properties of matter are related to its microscopic (invisible) composition and structure. The atoms and molecules that compose matter are too small to see. But if we know what something is made of, and how it's held together, then we can break it apart and rearrange it. The result can be something with totally different properties. Nylon is made from oil!

1.3 HOW CAN WE COMPARE MATERIALS?

Performance	**Physical & Chemical Properties**
Composition & Structure	**Processing & Synthesis**

These four categories are useful ways to sort different materials. Metals, polymers and ceramics tend to have great differences in these categories. Each category will be briefly discussed here, then used in later chapters to highlight the special qualities of each material.

PERFORMANCE

The **performance** of a material is discussed in the context of an application. For example, many materials are used for building houses. Once upon a time, three little pigs and a big bad wolf did an experiment to find out if a house should be made of straw, sticks or bricks.

Comparison of ceramic and polymer building materials

When the wolf huffed and puffed, the straw and stick houses fell down. Brick had the best performance. However in an earthquake, the brick house would be the worst place to be! The vibrations can rattle the bricks apart. California's building codes favor wood or steel-reinforced concrete.

What should a good house material do? Protect the things inside from weather - wind, cold, heat, rain, snow, hail - and from fire. A medieval castle was designed to withstand cannonballs but since that is not usually a concern in modern construction, most people would decide that its superior performance is not worth the extra cost.

PROPERTIES

To choose a material with the best performance for a particular application, we will need to consider the **properties** of the available materials. Properties are the observed characteristics of a sample.

Physical properties

Some physical properties describe how an object responds to mechanical forces. **Hardness** is one example of a mechanical property. If you drag a steel knife blade across a hard object, such as a plate, the hard surface is unchanged; if you drag the blade across a soft object, such as a piece of chalk, the soft surface will be scratched. An object is **tough** if force is unable to break or tear it. The response to force depends on the material's structure, and also on its shape and size. A piece of notebook paper can be torn easily, but a telephone book requires much greater force.

We can easily bend a **flexible** object such as a nylon jacket, but more force is required to bend a **stiff** object like a polyethylene milk jug. If an object returns to its original shape and size when the force is removed, we call it **elastic**. If the deformation remains, it is **plastic**. An object that breaks rather than bending is **brittle**.

An object is **strong** if an applied force is unable to deform or break it. A nylon windbreaker is strong, since pulling on it does not change its length. Sometimes the manner of applying a force makes a difference to the strength of an object. Ceramics can bear a lot of weight, but will break if stretched or bent. Nylon survives compression, pulling and twisting.

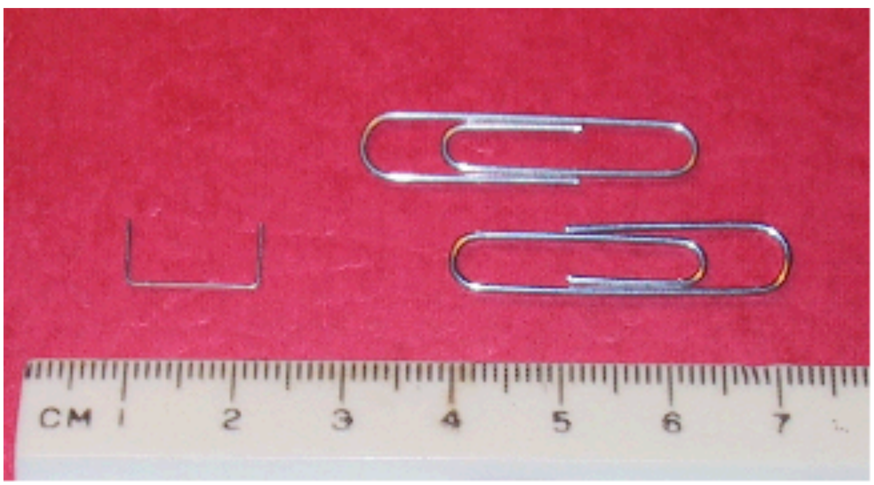
Color, **texture**, and **reflectivity** can be observed by shining light on a sample. Mirrors are colorless, smooth, and shiny. **Electrical conductivity** is detected by applying a voltage across an object. Applying heat to a sample reveals its ability to **conduct heat**, its **melting point** (temperature at which a solid changes to liquid), and its **boiling point** (temperature at which a liquid changes to gas).

Some properties are independent of the amount of sample. Melting point does not change if a sample is divided in half. Other properties, including **mass** and **volume**, increase with the amount of sample being studied.

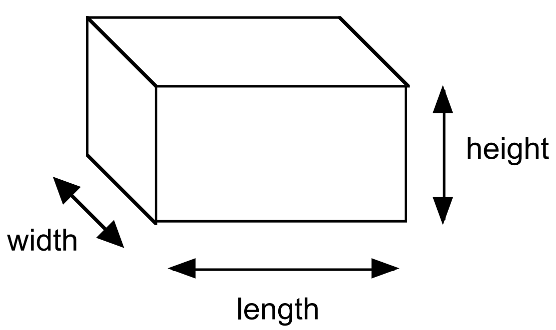
When observing properties in a laboratory, scientists use the Metric System of measurements. The basic units of the Metric System are presented in Appendix I.

Mass is measured on a balance or scale.

A ruler can be used to find the volume of a regularly shaped object.



A staple is just over 1 cm long.
Two paper clips have a mass of about 1 gram.



Volume = length x width x height

volume of liquid = 2.0 mL　　　　volume of liquid + object = 4.5 mL

volume of object = difference = 2.5 mL

Displacement of liquid in a graduated cylinder is a method that will work for irregularly shaped objects.

Some interesting properties are not measured directly, but can be determined by combining measurements of other properties. **Density** is a characteristic property of a material and can be a good way to identify an unknown sample. It is usually calculated from the mass and volume of a sample.

density = mass / volume　　　or　　　$d = m / V$

Example: A block of wood measures 2 cm by 1 cm by 5 cm. Its mass is 7.2 g.
What is its density?
The **volume** is 2 cm x 1 cm x 5 cm = 10 cm^3 = 10 mL.
The **density** is 7.2 g / 10 mL = **0.72 g/mL**

Most solids have densities greater than 1.0 g/mL, but there are exceptions; anything that floats on water (such as wood) has a density less than 1.0 g/mL. Although a chunk of alumimum will sink in water, aluminum foil can be shaped to create an object of greater volume that floats. The table below shows the range of densities for some common materials.

Densities of some common chemicals and materials

sample	density (g/mL)
cork	0.2
ethanol	0.8
water	1.0
rubber	1.1-1.2
salt	2.2
aluminum	2.7
cement	2.7-3.0
gold	19.3

Chemical properties

These describe what chemical reactions are likely to occur. We can observe how a sample reacts when mixed with other chemicals (water, acid). A material that can burn is described as flammable. Some materials rust (a type of oxidation reaction). Some materials dissolve in water or other liquids.

Usually a chemical reaction involves a transformation of the sample into a different substance, and it may be difficult to reverse the process. For example, wood is flammable. When it burns it combines with oxygen from the air. The reaction produces ashes, smoke and water; it cannot be reversed to make wood. The products of the reaction have quite different properties from the original wood.

Characteristic Properties of Major Classes

metals	polymers	ceramics
hard but malleable	stiff or flexible	hard but brittle
shiny	dull	shiny if glazed
little color	colorless	many colors
intermediate	low melting	highest melting point
conduct electricity	nonconductive	nonconductive
high density	low density	intermediate density
difficult to burn	flammable	not flammable

COMPOSITION

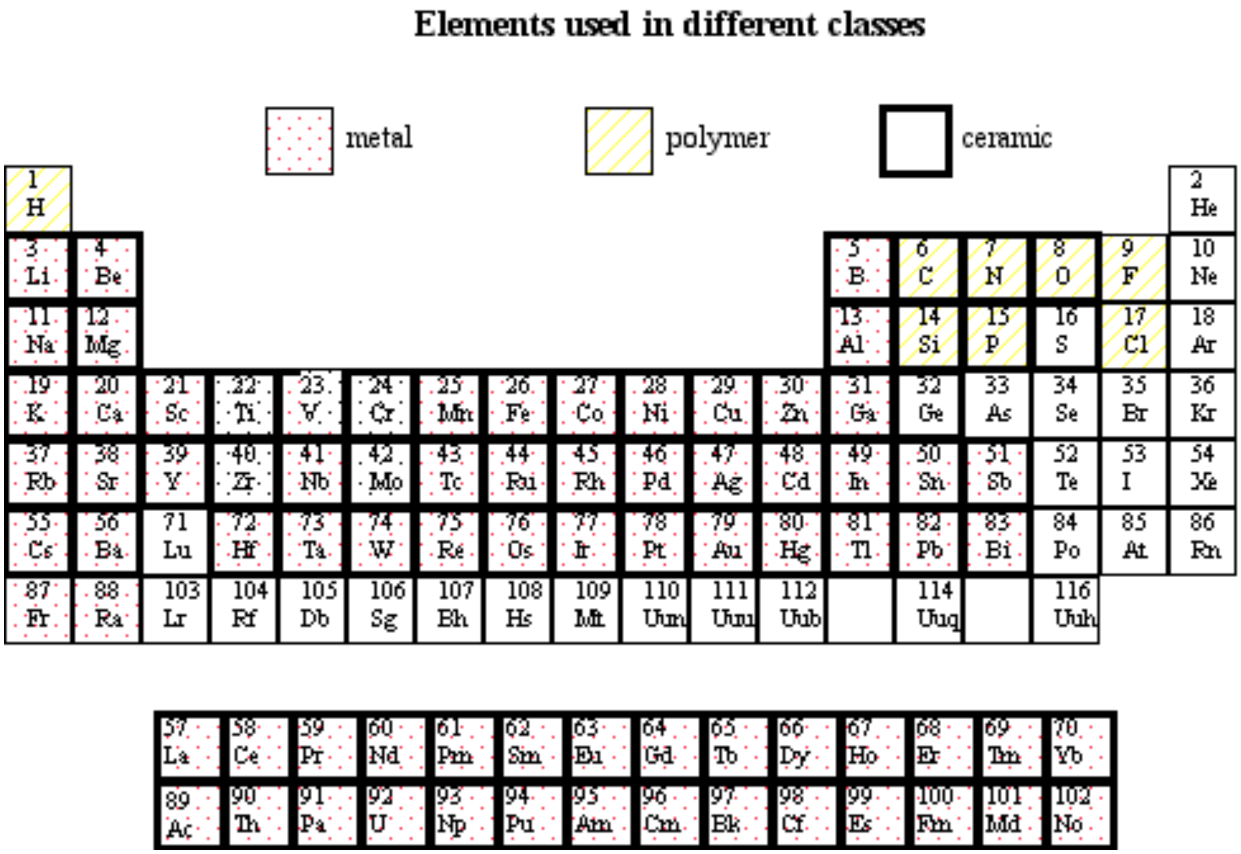
Composition tells what chemicals are in a sample. The most specific description will reveal the chemical elements that are present in the sample.

Chemists determined that matter is composed of combinations of about 100 elements. The simplest pieces chemists can make are atoms. Atoms of the same element are identical (have the same properties) and atoms of different elements are different. Some physical properties of elements are listed in Appendix II. Chemists can change oil into nylon only because they are composed of the same elements.

The Periodic Table of the Elements lists all the known elements. Each element's square has its atomic number, name, and one or two letter chemical symbol. For example,

1	hydrogen	H
8	oxygen	O

Many different experimental techniques have been developed to test which elements are present in a sample. One of the most sophisticated techniques, ***X-Ray Photoelectron Spectroscopy*** *(XPS), is described in Chapter 14. It is most often used to test metals and ceramics.*

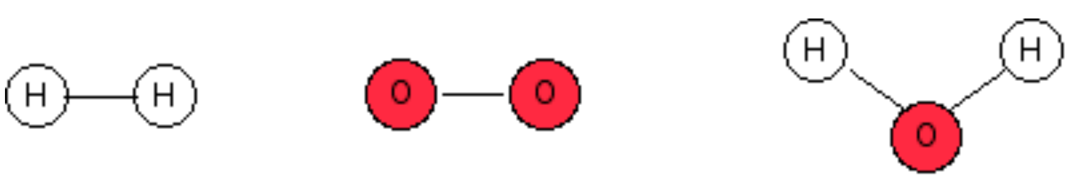


As shown in the figure, different sets of elements are found in metal, polymer and ceramic samples. The elements for metallic and polymeric materials do not overlap at all. This is one of the reasons that the properties of those two classes are so different. The elements found in ceramics are also in found metals or polymers.

Compounds

A compound has a uniform composition: every sample removed and tested, no matter how small, contains same elements in the same proportions. The elements in compounds are held together tightly and can only be separated from each other by chemical reactions.

A chemical formula relays the chemical composition of a compound in a compact manner. It includes the chemical symbol for each element, listed from left to right as they appear on the Periodic Table. The symbol for each element is immediately followed with the number of atoms of that element as a subscript. Fe_2O_3 contains two iron atoms and three oxygen atoms. If there is only one atom of the element, we do not bother to write the number one. Water has two hydrogens and one oxygen, so its chemical formula is written H_2O.



The elements hydrogen and oxygen, and the compound water.

Table salt is NaCl. Sand always contains one silicon for every two oxygens, so its formula is SiO_2.

Mixtures

If you mix sand and salt, any proportions of elements is possible. In a mixture, the different elements are not coordinated with each other. This means that samples from a mixture won't always have identical proportions. Bricks, steel, and concrete are materials that are mixtures.

The elements in a mixture are not tightly bound, so they often can be separated by physical processes. To separate a mixture of sand and salt you can add water since salt dissolves but sand does not. After straining out the sand, boil the water away from the salt.

Concrete is composed of cement, sand, and pebbles.

STRUCTURE

The three dimensional arrangement of atoms in a sample creates its **structure**. When sand changes into glass, its silicon and oxygen atoms shift positions to make a continuous sheet. Even though the chemical composition is the same, some of the properties have changed. Common structures for materials are described in Chapter Two.

PROCESSING AND SYNTHESIS

Various methods can be used to create materials from existing substances. Although *Star Trek* writers imagined a replicator which could produce any item of known composition and structure, real transformations are slower and messier.

Processing a material could be as simple as hammering a piece of copper, or flaking arrowheads from a piece of flint. When sand is melted and formed into glass, the primary change occurring is in the arrangement of the silicon and oxygen atoms. There are several ways to change iron ore, a compound, into iron metal. The different processes used to produce cast and wrought iron result in different ratios of iron and carbon which create different properties in the final objects, and make them suitable for different applications.

Synthesis implies a major change in chemical composition; for example, polymers are synthesized by cooking mixtures of chemicals. New molecular structures result.

1.4 HOW DOES ATOMIC STRUCTURE RELATE TO CHEMICAL PROPERTIES?

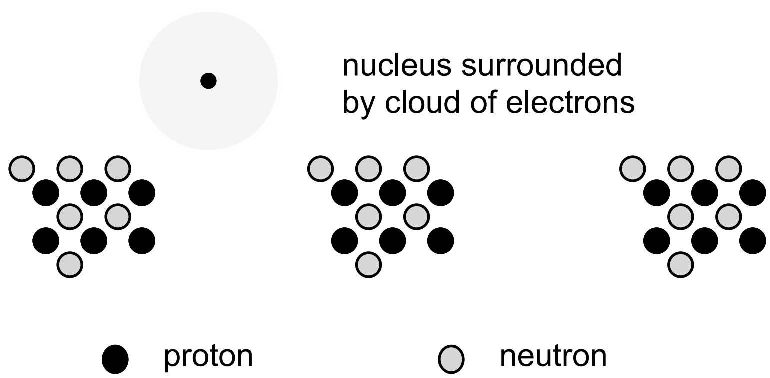
THE COMPONENTS OF AN ATOM

It has been mentioned that elements are the purest forms of chemicals. Chemical reactions cannot make them simpler. Each atom in a sample of an element has the same chemical properties.

In 1897 it was discovered that atoms can be separated into negatively charged and positively charged parts. **Electrons** are the pieces with negative charge. Chemists usually say that one electron has a charge of -1. Electrons can be rearranged by chemical reactions.

If all the electrons are removed from an atom, what remains is the **nucleus**. It is positively charged and contains most of mass of an atom. The positive charge of the nucleus attracts the negative charge of the electrons.

Further experiments showed that a nucleus is composed of two types of particles. Each **proton** in the nucleus has a charge of +1. A **neutron** has no charge. Both types of particle have much larger masses than electrons. Neutrons and protons are held together so tightly that an atomic nucleus will not change during a chemical reaction.



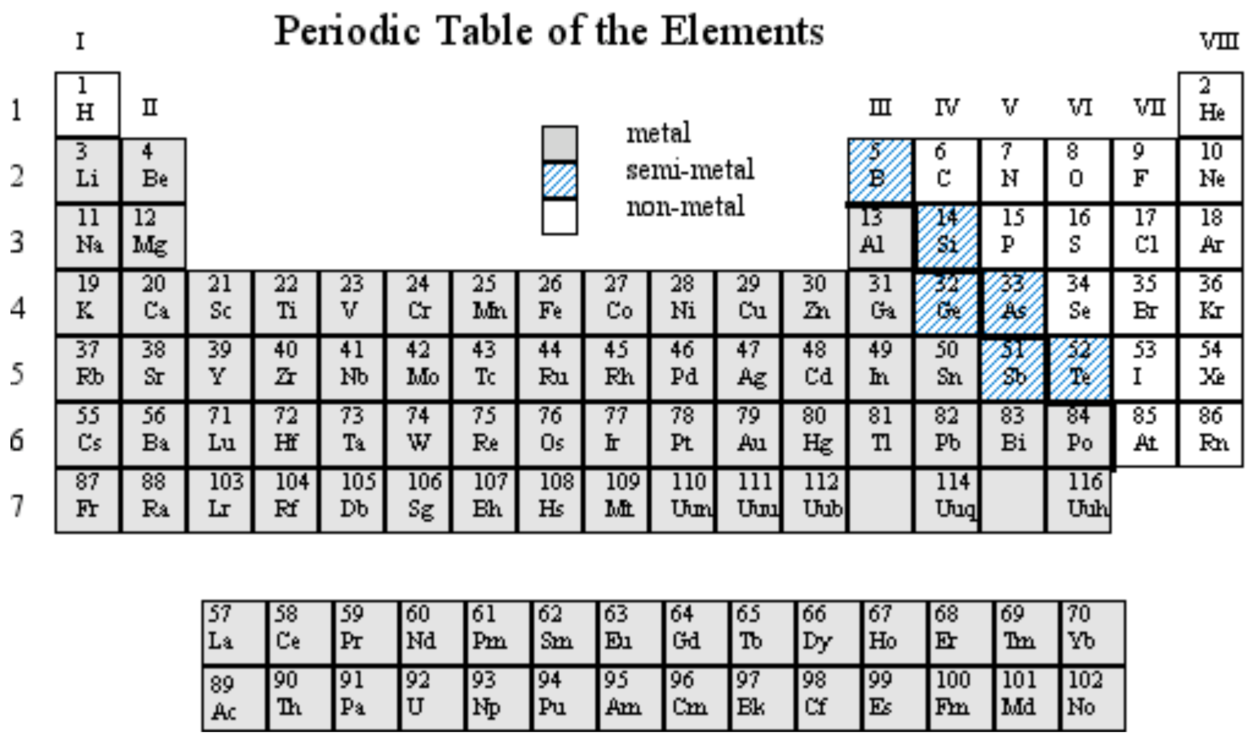
Carbon always has six protons in its nucleus, but it can have six, seven or eight neutrons.

Since an atom has no overall charge, its number of protons is equal to its number of electrons. A carbon atom always has six protons and six electrons.

Atoms are so small that 6.02×10^{23} carbon atoms are needed to make a 12 gram sample! The mass of an atom may be reported in atomic mass units (amu). 1 amu = 1.66×10^{-24} g.

ATOMIC NUMBER AND THE PERIODIC TABLE

The identity of an atom is determined by its number of protons. The atomic number found on the Periodic Table is the number of protons. Any atom with one proton is a hydrogen atom (with atomic number 1). Scientists have found atoms with as many as 116 protons in their nuclei, but the largest nuclei are unstable and not used for materials.



The order of elements on the table is the number of protons. Elements are similar to their left- and right-hand neighbors in mass and size. Elements farther down the periodic table show an increase in mass and size.

Notice that the rows labelled 1 through 7 (called "periods") are not all the same length. The pattern provides information about the chemical properties of elements. Elements in the same column (called a "group") undergo similar reactions. For example, group I contains the elements Li, Na, K which all react violently with water.

The tallest groups, labelled with Roman numerals, contain the **main group elements**. The central section of the table, starting with element #21, contains **transition elements**. The two rows at the bottom of the table contain **inner transition elements**.

Classes of elements

Left/bottom corner: **metals**. These are almost all solids at room temperature. They are shiny and conduct electricity well.

Right/top corner: **non-metals**. These are more often gas or liquid than solid; non-conductors. The last group on the right, labelled VIII, contains the noble gases. These almost never react and do not form useful materials.

Along the diagonal division: **semi-metals**. These have properties that are intermediate between those of metals and nonmetals. They are shiny, but too brittle to make wires. Their most useful property is that they are semiconductors. This means that they conduct electricity under controlled conditions.

Valence electrons

The chemical properties of an element are determined by number of electrons available for reactions. The theory called quantum mechanics explains that electrons are arranged in shells around each atomic nucleus. If a shell is completely filled, the electrons very stable. It is hard to remove or add any. Noble gases have filled shells, so rarely undergo chemical reactions. Elements from other groups will undergo reactions to gain or lose electrons.

The reactive electrons are called **valence electrons**. The number of valence electrons for a main group element is the same as its group number, I through VIII. If we know an element's valence, we can predict how it is likely to react to fill the shell: lose, gain, or share electrons. We will see later that the valence electron concept does not work as well for transition metals.

Ions

If energy is available, an atom can lose or gain electrons until it matches the closest noble gas (achieve a filled shell). The name of the atom is unchanged; since it has the same number of protons as before, it is still the same element. No chemical process can change the number of protons in a nucleus. However the positive charge of the nucleus is no longer exactly cancelled by negative electrons. An **ion** is a particle for which the number of protons is not equal to the number of electrons.

We write the chemical symbol with a superscript showing the ion's charge.

Example. Na has 1 valence electron; remove it to match Ne. Na^+ 11 p^+, 10 e-

Na could add 7 electrons to match Ar, but that large a change would require too much energy. In general, elements on the left of the periodic table (metals) form positive ions.

Elements on the right side of the periodic table (nonmetals) usually form negative ions.

Example. O has 6 valence electrons; we can add 2 to match Ne. O^{2-} 8 p^+, 10 e-

Losing 6 electrons to match He would be too difficult.

The most common ionic compound, table salt, has the chemical name sodium chloride. It contains equal numbers of sodium ions (Na^+) and chloride ions (Cl^-) and is known by the formula NaCl. No superscripts for charges appear in the compound's chemical formula because the total compound is neutral. The charges on the cations cancel out the charges on the anions. When we read the formula we must remember that a compound composed of a metal and a nonmetal is always ionic.

Learning Goals for Chapter 1

After studying this chapter you should be able to:

Recall historic eras labeled by materials.
Know the three major classes – Metals, Polymers, Ceramics.
Give examples of, and common applications for, each class of material.
Identify physical & chemical properties: what are they; which are common for each class of material.
Know how to find the density of a sample.
Use properties and composition to sort materials into classes.
Use a Periodic Table in the following ways:

- find numbers of protons, electrons, and atomic mass for an element;
- know what chemical relationships are found in groups and periods;
- locate regions of metals, semimetals, and nonmetals;
- predict stable ions of main group elements.

Use the metric system: know the basic units for length, volume, mass, and the prefixes.

Vocabulary list

atom	atomic mass	atomic number
ceramic	color	composition
conductivity	density	elastic
electron	element	flexible
hard	ion	mass
metal	neutron	nucleus
plastic	polymer	performance
processing	proton	reflectivity
stiff	strong	texture
tough	valence electron	volume